# Prior-Agnostic AFDM Channel Estimation for High-Mobility LEO Satellite Communications Using Diffusion Models

Yi Huang, Yuqing Guo, Ce Guo, Ying Wang, *Member, IEEE*, and Xufeng Guo

***Abstract*— Affine frequency division multiplexing (AFDM) is envisaged as a promising waveform for high-mobility low Earth orbit (LEO) satellite communications. This letter proposes a prior-agnostic AFDM channel estimation framework using score-based diffusion models. Specifically, the score function is learned directly on the AFDM effective channel in an unsupervised manner, and posterior inference is performed via likelihood-annealed Langevin dynamics using the AFDM observation model. Upon this approach, the proposed method avoids restrictive sparse or grid-aligned channel assumptions and is more robust to fractional delay-Doppler off-grid leakage. Simulations under standardized LEO channel models demonstrate that the proposed diffusion model-based method far outperforms conventional sparse recovery and representative learning-based baselines, while maintaining robustness across various parameter profiles.**

***Index Terms*— AFDM, LEO communications, channel estimation, likelihood-annealed Langevin dynamics, diffusion model.**

## I. Introduction

THE forthcoming sixth-generation (6G) wireless communications are envisaged to enable ubiquitous broadband access anytime and anywhere [1], catalyze the paradigm shift from terrestrial networks to non-terrestrial networks (NTN) where low Earth orbit (LEO) satellites are deployed to facilitate wide-area coverage and flexible access for remote or underserved regions [2]. However, the high orbital velocity of LEO satellites induces significant Doppler shifts, leading to time-frequency doubly selective channels under multipath propagation [3]. Such unfavorable fading severely undermines link reliability and induce strong inter-carrier interference (ICI) in conventional orthogonal frequency division multiplexing (OFDM) [4]. Therefore, novel waveforms that are suitable for high-mobility LEO environments are urgently required.

In recent years, various novel waveforms [5] have been proposed for high-mobility scenarios, such as orthogonal time-frequency-space (OTFS) [3], [6] and affine frequency division multiplexing (AFDM) [7]. Although both OTFS and AFDM can combat severe Doppler shifts, OTFS typically incurs relatively high computational complexity due to complicated two-dimensional (2D) inverse symplectic finite Fourier transform (ISFFT) [3]. In comparison, AFDM maps DAFT-domain symbols to the time–frequency domain via the one-dimensional IDAFT, which with properly designed chirp parameters provides strong Doppler robustness over doubly dispersive channels while incurring lower implementation complexity than OTFS, thus achieving better air-interface suitability [8].

Accurate acquisition of the DAFT-domain channel is crucial for AFDM implementations. Classical embedded-pilot estimators [9] often rely on idealized channel assumptions, while recent research advances typically adopt grid-based sparse recovery frameworks [10], [11], [12]. Score-based generative models, closely related to diffusion models, have exhibited promising potential for high-dimensional signal recovery by learning the score function of the target distribution and performing posterior inference via annealed Langevin dynamics [13], and have recently been applied to conventional MIMO channel estimation, while their use for AFDM channel estimation in high-mobility LEO scenarios remains largely unexplored.

This letter aims to develop a prior-agnostic AFDM channel estimation framework, which does not leverage any prior assumption on the channel. Specifically, we perform score-based learning and inference directly on the AFDM effective channel and incorporate the AFDM measurement model into likelihood-annealed updates, enabling high-fidelity estimation under practical LEO propagation environments. Moreover, the proposed framework is not merely restricted to the 3rd Generation Partnership Project (3GPP)-based channel realizations. The contributions of this letter can be summarized as follows.

***(i) Unsupervised Training:*** We apply a score model directly on the AFDM effective channel, and the training is unsupervised via denoising score matching, which does not require any explicit parametric prior, sparsity model, or statistical channel assumptions. ***(ii) Adaptive Inference:*** We develop an AFDM-aware inference procedure based on annealed Langevin dynamics, which is fundamentally a coarse-to-fine refinement that effectively balances observation fidelity and learned structural regularization. ***(iii) Simulation Validation:*** Simulation results under 3GPP standardized LEO channel models and practical noise levels validate the superiority of the proposed framework over classical linear estimators (e.g., LS), sparse-recovery baselines (e.g., SBL), and representative learning-based estimators (e.g., WGAN and L-DAMP), thereby demonstrating both precision gains and practical applicability.

## II. System Model

This section revisits the AFDM transmission principle and derives the input-output relationship of the AFDM system.

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Consider LEO uplink transmissions from a single-antenna ground terminal (Tx) to a single-antenna LEO satellite (Rx). Let $N$ denote the number of AFDM chirp subcarriers. Each AFDM frame spans a duration of $T = N\Delta t$. The sampling rate is given by $1/\Delta t$, and the subcarrier spacing to $\Delta f = 1/T$. The DAFT-domain symbol vector is mapped to the time domain via an $N$-point inverse IDAFT [14], which can be compactly expressed in matrix form as $\mathbf{s} = \mathbf{A}^H\mathbf{x} = \mathbf{\Lambda}_{c_1}^H\mathbf{F}^H\mathbf{\Lambda}_{c_2}^H\mathbf{x}$, where $\mathbf{A}$ denotes the DAFT matrix, $c_1$ and $c_2$ are the AFDM chirp-design parameters, $\mathbf{\Lambda}_c$ is a diagonal chirp-modulation matrix, and $\mathbf{F}$ is the unitary $N$-point DFT matrix.

The ground-space LEO channel can be characterized as a sparse multipath channel in the delay-Doppler (DD) domain

$$h(\tau,\nu) = \sum_{i=0}^{P-1} h_i \delta(\tau-\tau_i)\,\delta(\nu-\nu_i), \tag{1}$$

where $h_i$ denotes the complex channel coefficient associated with the $i$-th path, $\delta(\cdot)$ represents the Dirac delta function, $\tau_i \in [0,\tau_{\max}]$ and $\nu_i \in [-\nu_{\max},\nu_{\max}]$ denote the delay and Doppler shift of the $i$-th path, respectively, which can be expressed as $\tau_i = \frac{l_{\tau_i}}{N\Delta f}$, $\nu_i = \frac{k_{\nu_i}}{T}$, where $l_{\tau_i} = \bar{l}_\tau^i + \iota_i$ and $k_{\nu_i} = \bar{k}_\nu^i + \kappa_i$ denote the delay and Doppler indices of the $i$-th path, respectively, with $l_{\tau_i} \in [0, l_{\max}]$ and $k_{\nu_i} \in [-k_{\max}, k_{\max}]$. The parameters $\tau_{\max}$ and $\nu_{\max}$ are determined by the maximum propagation distance and the largest relative velocity between the transceivers and scatterers, respectively. Moreover, $\bar{l}_\tau^i$ and $\bar{k}_\nu^i$ are the integer components of the delay and Doppler indices, whereas $-\frac{1}{2} < \iota_i, \kappa_i \leq \frac{1}{2}$ capture the corresponding fractional offsets from the nearest integer indices. For LEO satellite communications, the path delay, Doppler shift, and complex gain are specified as follows.

*Doppler shift:* Take the ground user as the stationary reference point. Let $v_s$ represent the velocity of the satellite, and the corresponding Doppler shift can be expressed as

$$\mathcal{D}(\psi) = \frac{f_c}{c} v_s \cos\psi, \tag{2}$$

where $\psi$ denotes the angle between the satellite velocity vector and the satellite-user line of sight, $f_c$ is the carrier frequency, and $c = 3\times10^8$ m/s is the speed of light. The resulting Doppler term $\mathcal{D}(\psi)$ is assumed to be common to all propagation paths within the same satellite-to-ground (S2G) link [15]. This is because, in LEO S2G scenarios, inter-path Doppler differences mainly arise from path-dependent Doppler angles, while the substantially large satellite-to-ground distance leads to only a small angular spread across multipath components.

*Delay:* We normalize and scale the path delays in the LEO channel as [16, Eq. (2)]. This process can be expressed as

$$\tau_i = \tau_{i,\text{norm}} \cdot \tau_{\text{spread}}, \quad \forall i \in \{0,\ldots,P-1\}, \tag{3}$$

where $P$ (an unknown variable for channel estimation) is the number of propagation paths due to the random and unpredictable scattering environment. $\tau_{i,\text{norm}}$ is the normalized delay of the $i$th path specified in 3GPP TR 38.811 [17] (see **Table** I), $\tau_{\text{spread}}$ represents the prescribed delay spread.

*Channel Gain:* We employ the NTN-TDL-D profile [17], which consists of 3 taps—1 Rician and 2 Rayleigh—whose parameters are listed in **Table** I. After propagation over the terrestrial-satellite link (TSL), the Rx observes a noisy superposition of delayed and Doppler-shifted transmitted waveforms. The AFDM receiver applies DAFT to transform the time-domain samples $r[n]$ back to the DAFT domain, yielding the observation symbols $y[m]$ for subsequent processing. This can be characterized by matrix-form input-output relation as

TABLE I
NTN-TDL-D CHANNEL MODEL [17, TABLE. 6.9.2-4]

| Tap | Normalized Delay | Power (dB) | Fading distribution |
|---|---|---|---|
| 1 | 0 | -0.284 | LOS path, Ricean |
| | 0 | -11.991 | NLOS path, Rayleigh |
| 2 | 0.5596 | -9.887 | NLOS path, Rayleigh |
| 3 | 7.3340 | -16.771 | NLOS path, Rayleigh |

$$\mathbf{y} = \mathbf{H}_{\text{eff}}\mathbf{x} + \tilde{\mathbf{z}} = \sum_{i=1}^{P} h_i \mathbf{H}_i \mathbf{x} + \mathbf{A}\mathbf{z}, \tag{4}$$

where $\mathbf{H}_{\text{eff}}$ denotes the effective channel matrix. Since $\mathbf{A}$ is unitary, noise $\mathbf{z}$ and $\tilde{\mathbf{z}}$ are statistically identical [14], each of which follows a zero-mean circularly symmetric complex Gaussian distribution with variance $\sigma_z^2$. To better capture the LEO channel dynamics, we incorporate fractional delay and fractional Doppler effects, and the resulting expression for the elements of the channel matrix $\mathbf{H}_i$ is given as follows

$$H_i[m,m'] = \frac{1}{N} \underbrace{e^{j2\pi\left(c_1 l_{\tau_i}^2 - c_2\left(m^2-m'^2\right) - \frac{l_{\tau_i} m'}{N}\right)}}_{\mathcal{P}_i(m,m')} \times \underbrace{\sum_{q=0}^{C} e^{j2\pi q\,\iota_i} \sum_{(n-l_{\tau_i})_N \in \boldsymbol{\mathcal{L}}_{m,q}} e^{j\frac{2\pi}{N}\left(m'-(m+l_i^{\text{eq}})\right)n}}_{\mathcal{F}_i(m,m')}, \tag{5}$$

where the equivalent delay is defined as $l_i^{eq} \triangleq k_{\nu_i} + C l_{\tau_i}$, $C \triangleq 2c_1 N$ and $\boldsymbol{\mathcal{L}}_{m,q} \triangleq [n_{m,q}+1,\ n_{m,q+1}]$, with $n_{m,q} \triangleq \left\lfloor \frac{t_{m,q}}{\Delta t} \right\rfloor$.

In practical propagation environments, fractional Doppler shifts and fractional time delays can introduce significant power leakage in the effective channel matrix of (5), thereby weakening the sparsity of $\mathbf{H}_{\text{eff}}$. Consequently, conventional compressive sensing (CS)-based estimators [3], [18], [19] may incur severe performance degradation in the presence of off-grid effects. To address this issue, we develop a diffusion framework that directly learns the distribution of the AFDM effective channel, enabling precise estimation of off-grid $\mathbf{H}_{\text{eff}}$.

## III. SCORE-BASED GENERATIVE CHANNEL ESTIMATION

The proposed score-based generative model-driven channel estimation scheme consists of two stages: a training stage and an inference stage. During the training stage, we train a diffusion model $s_\theta$ by minimizing a loss function. During the inference stage, we iteratively refine the effective AFDM channel estimate using $s_\theta$ aided by the pilot signals.

### A. Score-Based Model Training for AFDM Channel

In the standard pilot-aided channel estimation procedure, we exploit multiple pilot transmissions for channel estima-

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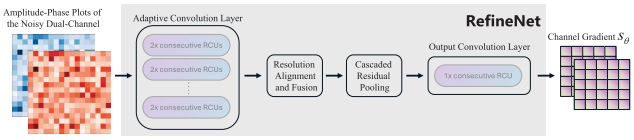



Fig. 1. Detailed architecture of $s_\theta$ based on RefineNet.

tion [9]

$$\mathbf{Y} = \mathbf{H}_{\text{eff}}\mathbf{X} + \tilde{\mathbf{Z}}, \tag{6}$$

where $\mathbf{X}$ and $\tilde{\mathbf{Z}} \in \mathbb{C}^{N\times N_P}$ denote the transmitted pilot symbol matrix and the noise matrix, respectively. $\mathbf{Y}$ denotes the received pilot symbol matrix. Each entry of the pilot matrix $\mathbf{X}$ is randomly drawn from the quaternary phase shift keying (QPSK) constellation $\{1+j, 1-j, -1+j, -1-j\}$.

Conventional AFDM pilot designs usually rely on the DD-domain sparsity of the channel, so that a single embedded pilot sequence is sufficient to estimate the small set of channel parameters. Since we directly estimate the full AFDM effective channel matrix $\mathbf{H}_{\text{eff}}$ without imposing sparsity assumptions, using only one pilot sequence makes (6) underdetermined, so that multiple pilot sequences are required. Meanwhile, although AFDM can only guarantee approximate channel invariance within a finite signaling duration, the effective channel variation across several adjacent pilot sequences remains negligible because the corresponding Doppler-angle variation over a short interval is relatively small within pilot frames.

In practice, we parameterize the score function $s_\theta$ using a deep RefineNet [20] architecture with learnable weights $\theta$ as Fig. 1. RefineNet is a multi-path refinement network consisting of several cascaded blocks with residual skip connections, where each block fuses multi-resolution features via residual convolution units and a multi-resolution fusion module.

The training objective for $s_\theta$ with $B$ samples is [13]

$$\mathcal{L}(\theta) = \frac{1}{B}\sum_{i=1}^{B}\sigma_i^2\left\| s_\theta\left(\mathbf{H}_{\text{eff}}^{(i)} + \mathbf{N}_i\right) + \frac{\mathbf{N}_i}{\sigma_i^2}\right\|_2^2, \tag{7}$$

where $\mathbf{N}_i \sim \mathcal{CN}\left(\mathbf{0}, \sigma_i^2\mathbf{I}\right)$ with training noise variances $\sigma_i^2$.

Since the AFDM effective channel matrix is complex-valued, we adopt a two-channel real–imaginary representation to enable the real-valued RefineNet parameterization. In each mini-batch, we corrupt the clean channel realizations $\mathbf{H}_{\text{eff}}^{(i)}$ from the training set by adding complex Gaussian noise at the specific level. Then, we train the model to predict the scaled negative noise term $-\frac{\mathbf{N}_i}{\sigma_i^2}$, where the network output is interpreted as a two-channel score associated with the gradients with respect to the real and imaginary components. As a result, the resulting inference is more robust and can more readily approach the theoretical Cramer-Rao lower bound.

Although RefineNet is a generic score-model backbone, our framework is tailored to AFDM channel estimation by directly estimating $\mathbf{H}_{\text{eff}}$ in the AFDM physical domain, thereby avoiding gridded sparse-domain mismatch and power-leakage loss.

### B. Posterior-Sampling-Based Inference for AFDM Channel

After obtaining the trained score-based AFDM channel model, we perform inference by iteratively refining the channel estimate via posterior sampling of the channel realizations which is also tailored to AFDM channel estimation. At each

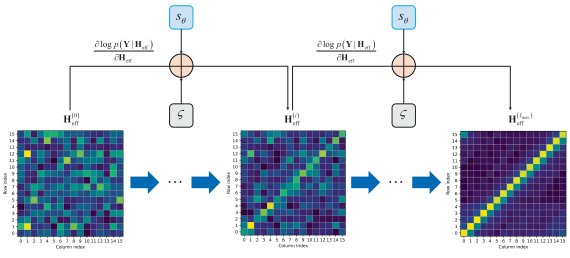



Fig. 2. Iterative Inference via a Score-Based Model $s_\theta$.

iteration, we employ an annealed Langevin dynamics-based iterative algorithm [13] to update the refined estimate

$$\mathbf{H}_{\text{eff}}^{(i+1)} = \mathbf{H}_{\text{eff}}^{(i)} + \alpha_i\nabla\log p\left(\mathbf{H}_{\text{eff}}^{(i)}|\mathbf{Y}\right) + \sqrt{2\beta\alpha_i}\sigma_i\varsigma, \tag{8}$$

where $\varsigma \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$ aims to generate perturbation and avoid falling into a local minimum, which is newly sampled in each iteration. Note that the parameters $\alpha_0$, $\beta$, and $r$ are pre-specified hyper-parameters in the refinement process. According to Bayes rule, $\nabla\log p\left(\mathbf{H}_{\text{eff}}^{(i)}|\mathbf{Y}\right)$ in (8) is recast to

$$\nabla\log p\left(\mathbf{H}_{\text{eff}}^{(i)}|\mathbf{Y}\right) = \nabla\log p\left(\mathbf{Y}|\mathbf{H}_{\text{eff}}^{(i)}\right) + \nabla\log p\left(\mathbf{H}_{\text{eff}}^{(i)}\right), \tag{9}$$

where $\nabla\log p\left(\mathbf{Y} \mid \mathbf{H}_{\text{eff}}^{(i)}\right)$ is the likelihood score, which enforces consistency with the observation, and $\nabla\log p\left(\mathbf{H}_{\text{eff}}^{(i)}\right)$ is the prior score, which encourages the estimate to conform to the channel distribution learned during training and can be approximated by the trained network $s_\theta$. Then, the corresponding likelihood score can be further written in the form as [13]

$$\nabla\log p\left(\mathbf{Y}|\mathbf{H}_{\text{eff}}^{(i)}\right) = \frac{\left(\mathbf{H}_{\text{eff}}^{(i)}\mathbf{X} - \mathbf{Y}\right)\mathbf{X}^{\text{H}}}{\sigma_z^2}. \tag{10}$$

The score-based inference procedure is summarized in **Algorithm 1**. We adopt an annealing strategy by introducing noise levels $\{\sigma_i\}_{i=0}^{I_{\max}}$ to construct a coarse-to-fine inference procedure. This design prevents overly aggressive data-driven updates at large $\sigma_i$ and naturally reduces to the standard Gaussian likelihood weighting $1/\sigma_z^2$ as $\sigma_i \to 0$. Moreover, we employ a decaying step-size schedule $\alpha_i = \alpha_0 r^i$ $(0 < r < 1)$, so that larger steps at high noise levels facilitate global exploration. Specifically, $\beta$ scales the stochastic perturbation in the Langevin update. The parameter $r \in (0,1)$ specifies the geometrically decaying step-size schedule $\alpha_i = \alpha_0 r^i$ across the annealed noise levels. The algorithm performs $\boldsymbol{I}_{\max}$ outer iterations over the annealed noise levels $\{\sigma_i^2\}$, and at each noise level it executes $M$ inner updates with $\sigma_i^2$ held fixed. The initialization $\mathbf{H}_{\text{eff}}^{(0)}$ is adopted because Gaussian noise provides a mathematically consistent and non-informative initialization for the subsequent annealed Langevin refinement. The overall iterative inference procedure is outlined and depicted in Fig. 2.

## IV. Simulation Results

Simulation results are presented to evaluate performance of our proposed diffusion model-based channel estimation methodology. The AFDM system employs $N = 16$ DAFT-domain symbols with a carrier frequency of 2 GHz and a chirp subcarrier spacing of $\Delta f = 30$ kHz [9]. The AFDM

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**Algorithm 1** Score-Based AFDM Channel Estimation

**Initialization:**The trained network $s_\theta$, the transmitted pilot matrix $\mathbf{X}$, the received matrix $\mathbf{Y}$, the initial effective channel estimate $\mathbf{H}_{\text{eff}}^{(0)} \sim \mathcal{CN}(0, \mathbf{I})$, the noise variance $\sigma_z^2$, the number of noise levels $\boldsymbol{I}_{\max}$, the annealed noise level $\sigma_i^2$, the number of inner iterations $M$, and the hyper-parameters $\alpha_0$, $\beta$, and $r$, the convergence tolerance $\varepsilon$.

1: **while** $i \leq \boldsymbol{I}_{\max}$ and $\frac{\left\|\mathbf{H}_{\text{eff}}^{(i+1)} - \mathbf{H}_{\text{eff}}^{(i)}\right\|_2}{\left\|\mathbf{H}_{\text{eff}}^{(i)}\right\|_2} \geq \varepsilon$ **do**

2: **for** $m = 1$ to $M$ **do**

3: $\mathbf{H}_{\text{eff}} \leftarrow \mathbf{H}_{\text{eff}} + \frac{\alpha_0 r^i (\mathbf{H}_{\text{eff}}\mathbf{X} - \mathbf{Y})\mathbf{X}^{\text{H}}}{\sigma_z^2 + \sigma_i^2} + \alpha_0 r^i s_\theta(\mathbf{H}_{\text{eff}}) + \sqrt{2\beta\alpha_0 r^i}\sigma_i \varsigma$.

4: **end for**

5: $i = i + 1$.

6: **end while**

**Output:**The AFDM effective channel matrix $\tilde{\mathbf{H}}_{\text{eff}}$.

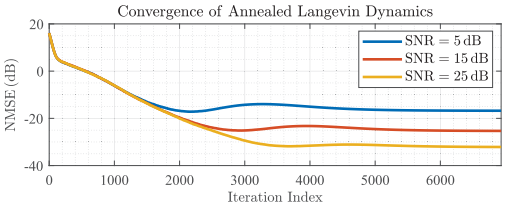



Fig. 3. Convergence Trajectories of Annealed Langevin Dynamics.

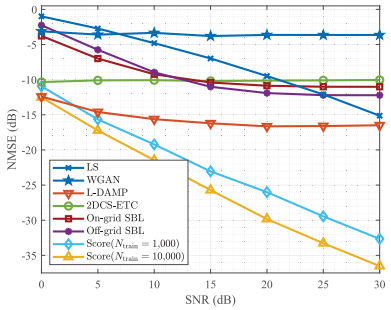



Fig. 4. Performance comparison of the score-based method with different TDL models versus SNR.

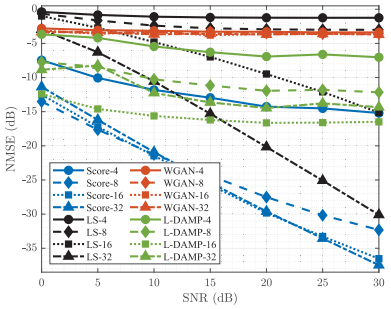



Fig. 5. Performance comparison of the score-based and other schemes with different $N_p$ versus SNR.

parameters are set to $c_1 = \frac{2(k_{\max}+k_v)+1}{2N}$ and $c_2 = \frac{1}{20N}$, and the frame duration is $T = 1/\Delta f$. The signal-to-noise ratio (SNR) is defined as $1/N_0$ and varied from 0 to 30 dB. The termination threshold of **Algorithm 1** is set to $\varepsilon = 10^{-3}$.

The ground-to-space link is modeled based on the 3GPP NTN-TDL-D profile summarized in **Table I**. The satellite operates at an orbital altitude of $H_0 = 500$ km and serves a ground footprint determined by the minimum elevation angle $\alpha = 25°$ required for link establishment. The satellite velocity

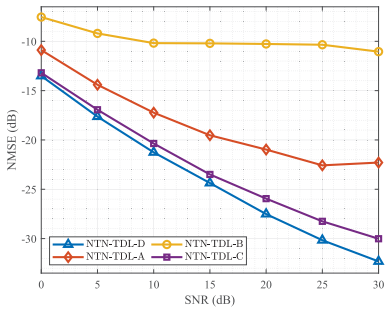



Fig. 6. Performance comparison of the score-based method with different TDL models versus SNR.

TABLE II

COMPUTATIONAL COMPLEXITY COMPARISON. NOTE THAT THE SCORE-BASED METHOD CAN BE ACCELERATED USING GPU

| Method | FLOPs (GFLOPs) | Order | Lat. (ms) | Mem. (MB) | Size (kB) |
|---|---|---|---|---|---|
| LS | $1.53 \times 10^{-4}$ | $O(N_p N^2 + N^3)$ | 0.89 | 9.93 | n/a |
| WGAN | $2.28 \times 10^{2}$ | $O(T_{\text{wgan}}(C_{\text{net}} + N_p N^2))$ | 7717.56 | 18.72 | 1156.01 |
| L-DAMP | 1.04 | $O(T_{\text{l}}(C_{\text{den}} + N_p N^2))$ | 67.61 | 18.48 | 8357.58 |
| 2DCS-ETC | $4.90 \times 10^{-5}$ | $O(T_{\text{cs}}(N_{\text{obs}} P + P^3))$ | 127.02 | $O(N_{\text{obs}} P + N_{\text{obs}}^2)$ | n/a |
| Ongrid-SBL | $1.89 \times 10^{-3}$ | $O(T_{\text{sbl}}(N_{\text{obs}} P^2 + P^3))$ | 1473.95 | $O(P^2 + N_{\text{obs}} P)$ | n/a |
| Offgrid-SBL | $2.00 \times 10^{-2}$ | $O(T_{\text{sbl}}(N_{\text{obs}} P^2 + P^3))$ | 6627.35 | $O(P^2 + N_{\text{obs}} P)$ | n/a |
| Score-based | $1.42 \times 10^{3}$ | $O(K(C_{\text{net}} + N_p N^2))$ | 158270.29 | 42.83 | 23008.13 |

is computed as $v = \sqrt{GM/r}$. The normalized maximum Doppler and delay spreads are set to $k_{\max} = 2$ and $l_{\max} = 1$, respectively. The path delays are generated according to (3) with the delay spread selected from $[30, 50]$ ns. All paths are assumed to share a common Doppler shift, determined by the satellite velocity and the satellite-UE geometry. The complex path gains are generated based on 3GPP TR 38.811 [17].

During the training process of the diffusion model, we randomly generate $N_{\text{train}} = 10,000$ AFDM channel matrices based on the NTN-TDL-D model [17] to form the training set, which can capture channel variations under different elevation angles. In constructing the test set, we randomly generate 100 AFDM channel matrices based on each of the NTN-TDL-A/B/C/D models to evaluate the robustness of the proposed score-based model. The following baseline schemes are adopted for performance comparison in our simulations. **LS:** The least squares algorithm in [21]; **WGAN:** The Wasserstein generative adversarial network in [22]; **L-DAMP:** The learned denoising-based approximate message passing algorithm in [23]; **2DCS-ETC:** The 2DCS-ETC algorithm in [24]; **1D on-grid SBL:** The 1D on-grid SBL algorithm in [19]; **1D off-grid SBL**: The 1D off-grid SBL algorithm in [25]. Table II compares the per-sample, per-estimation complexity of the score-based method with other baselines under $N = 16$ and SNR$= 15$ dB. In contrast, the score-based method scales as $O\big(K(C_{\text{net}} + N_p N^2)\big)$ with $K$ network evaluations, leading to higher total FLOPs/latency in this setting while achieving a substantially improved estimation accuracy. We use the normalized mean square error (NMSE) to evaluate and compare the channel estimation accuracy of the considered algorithms. The NMSE is defined as $\text{NMSE} = \frac{\left\|\mathbf{H}_{\text{eff}} - \tilde{\mathbf{H}}_{\text{eff}}\right\|_2^2}{\left\|\mathbf{H}_{\text{eff}}\right\|_2^2}$.

Fig. 3 depicts the convergence behavior of the Langevin dynamics across various iteration steps and SNR levels. It can be observed that, for all SNR levels, the annealed Langevin dynamics is able to achieve a stable convergence behavior.

Fig. 4 presents a performance comparison between our proposed diffusion model-based AFDM channel estimation scheme and six baselines. It can be observed that the proposed

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diffusion model-based method consistently outperforms the conventional schemes across the entire SNR range. Moreover, the CS approaches encounter a performance bottleneck when SNR $>$ 20 dB. This is because the CS approaches suffer severely from the power leakage effect due to the off-grid misalignment between the precise DD parameters and on-grid DD parameters, indicating a weak sparsity of the effective channel. In contrast, our proposed channel estimation framework does not rely on the strict sparsity of the effective channel but learns the channel characteristics from the training set, the estimation performance of which is not restricted by the power leakage effect caused by off-grid misalignment. WGAN and L-DAMP exhibit limited model capacity and poor generalization and their training is often unstable and highly sensitive to hyperparameters. As a result, they typically require retraining or extensive tuning across different operating points, leading to low engineering practicality and deployment efficiency. Notably, a similar performance advantage of the score-based method is maintained even when the training set size is reduced to 1,000 samples. With a sufficiently large training set, the score-based channel estimation method can approximately approach the theoretical error lower bound.

Fig. 5 illustrates the impact of the pilot length $N_p$ on the performance of our proposed diffusion model-based AFDM channel estimation and other methods. For a fixed symbol number $N$, we change the pilot length as $\{0.25N, 0.5N, N, 2N\}$ for performance comparison, corresponding to $N_p = 4, 8, 16$ and 32, respectively. It can be observed that when $N_p \geq N$, the performance of the diffusion model-based channel estimation method exhibits merely marginal improvement. When $N_p = 0.5N$, the performance is close to its converged level, whereas further reducing the pilot number leads to a pronounced performance degradation. Therefore, to balance computational complexity, estimation accuracy, and pilot overhead, choosing the pilot number as $N_p = 0.5N$ is the most suitable.

Fig. 4 compares the performance of the model trained under the NTN-TDL-D profile when evaluated across all TDL channel models. It can be observed that although the training set only contains channel samples generated from the NTN-TDL-D model, the proposed diffusion model-based estimation scheme remains a certain degree of robustness across different TDL profiles. Compared with the other profiles, the performance under NTN-TDL-B is relatively worse, which can be attributed to its larger mismatch in channel parameters with respect to NTN-TDL-D (the training set is generated by NTN-TDL-D). In future work, incorporating more TDL models and real-world channel measurement results into the training set can further enhance the robustness of our proposed method.

## V. Conclusion and Future Research Direction

This letter proposed a prior-agnostic, score-based inference for AFDM channel estimation in LEO communications. Operating directly on the AFDM effective channel, the proposed framework avoids restrictive channel assumptions and is robust under practical doubly dispersive conditions. Future research will focus on developing lower-complexity network training and inference strategies to further improve the practical efficiency of the proposed methodology. Another alternative direction is to move toward real LEO air-interface implementations, including deployment-oriented algorithm design and validation under realistic hardware and wireless propagation conditions.

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